

Recognition of Zhejiang License Plates Based on HSV Color Segmentation and SVM Character Classification

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Course: Machine Vision and Image Processing

Abstract: A machine-vision-based license plate recognition method is designed and implemented for Zhejiang blue license plates. To make the system suitable for a course project and easy to reproduce, the proposed method combines traditional image processing with support vector machine (SVM) character classification. First, the input vehicle image is converted from RGB/BGR to HSV color space, and the blue plate region is extracted by color thresholding. Morphological closing and contour filtering are then used to locate the license plate according to its area and aspect ratio. After plate cropping, grayscale conversion, Gaussian filtering, Otsu binarization, border removal and contour-based character segmentation are performed. The first Chinese provincial character is fixed as Zhejiang, while the remaining six alphanumeric characters are normalized to 20×20 binary images and recognized by a PCA-SVM model. Synthetic character samples and augmented real segmented characters are combined to improve recognition under the actual plate font. Experiments on two Zhejiang license plate images, Zhe A809JC and Zhe B2FS80, show that the proposed pipeline can correctly locate, segment and recognize the tested plates.

Keywords: machine vision; license plate recognition; HSV color segmentation; morphological processing; support vector machine; character recognition

0 Introduction

Automatic license plate recognition (ALPR) is an important application of machine vision in intelligent transportation, parking management and vehicle monitoring. A complete ALPR system generally includes three stages: license plate localization, character segmentation and character recognition [8, 1]. Although deep learning methods have been widely used in recent years, traditional image processing methods are still valuable in small-scale engineering experiments because they are interpretable, lightweight and easy to implement with OpenCV.

In practical vehicle images, license plate recognition may be affected by vehicle body texture, illumination variation, plate inclination, screw noise and character deformation. For Chinese blue license plates, the blue background provides a useful color cue. Therefore, this project uses HSV color segmentation to extract blue regions and then applies morphological processing and contour filtering to locate the plate. This strategy can reduce false detections from headlights, vehicle grilles and other horizontal edges.

The reference paper provided by the instructor presents a typical machine-vision research structure, including image acquisition, color-space processing, threshold segmentation, morphological processing, edge detection and experimental validation

[9]. Following the same engineering logic, this paper presents the system workflow, key image processing algorithms and experimental results for a Zhejiang license plate recognition task. Considering the limited scale of the course project and the difficulty of Chinese character recognition, the research object is limited to Zhejiang blue license plates. The first provincial character is fixed as Zhejiang, while the following six alphanumeric characters are automatically recognized.

1 Principle and Method

1.1 Experimental Environment and System Workflow

The project was implemented in PyCharm Professional on Windows. The programming language was Python. The main libraries included OpenCV for image processing, NumPy for numerical operations, scikit-learn for SVM training, joblib for model saving, and Pillow for generating synthetic character samples. The tested OpenCV version was 4.13.0. The input images were ordinary vehicle photographs containing blue Zhejiang plates.

The overall workflow is shown in Fig. 1. The system first extracts the blue plate region in HSV color space. The candidate plate area is selected using morphological closing, contour detection and aspect-ratio

constraints. After the plate is cropped, the characters are binarized, segmented, normalized and sent to the trained PCA-SVM classifier. The final recognition result is obtained by concatenating the fixed provincial prefix with the six recognized alphanumeric characters.

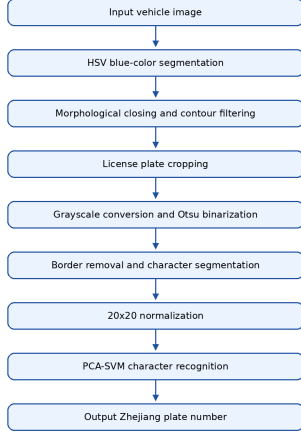


Fig. 1 Overall workflow of the proposed license plate recognition system

1.2 HSV Color Segmentation

The original vehicle image is first converted from BGR/RGB color space to HSV color space. Compared with the RGB model, the HSV model separates chromatic information from brightness information. This property is useful for blue license plate extraction because the hue component is relatively stable under moderate illumination variation.

For a pixel with normalized RGB values R , G and B , the maximum and minimum components are denoted by

$$C_{\max} = \max(R, G, B), \quad C_{\min} = \min(R, G, B), \quad (1)$$

and the chromatic difference is

$$\Delta = C_{\max} - C_{\min}. \quad (2)$$

The HSV components can be expressed as

$$V = C_{\max}, \quad (3)$$

$$S = \begin{cases} 0, & C_{\max} = 0, \\ \frac{\Delta}{C_{\max}}, & C_{\max} \neq 0, \end{cases} \quad (4)$$

and

$$H = \begin{cases} 60^\circ \times \frac{G - B}{\Delta}, & C_{\max} = R, \\ 60^\circ \times \left(2 + \frac{B - R}{\Delta}\right), & C_{\max} = G, \\ 60^\circ \times \left(4 + \frac{R - G}{\Delta}\right), & C_{\max} = B. \end{cases} \quad (5)$$

If the calculated hue is negative, 360° is added to keep it within $[0^\circ, 360^\circ]$.

Since the target license plates in this project are blue plates, the HSV image is thresholded according to a predefined blue range. In the OpenCV implementation, the hue value is scaled to $[0, 180]$, and the selected blue range is approximately

$$95 \leq H \leq 145, \quad S \geq 50, \quad V \geq 50. \quad (6)$$

The binary mask $M(x, y)$ is defined as

$$M(x, y) = \begin{cases} 255, & H_l \leq H(x, y) \leq H_u, \\ & S(x, y) \geq S_l, \quad V(x, y) \geq V_l, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

where H_l and H_u are the lower and upper hue thresholds, and S_l and V_l are the lower thresholds for saturation and value. This step converts the vehicle image into a binary map in which the blue license plate region is highlighted and most non-blue background regions are suppressed.

1.3 Morphological Processing and Contour Filtering

After HSV threshold segmentation, the blue mask may contain holes, discontinuities and isolated noise pixels. Morphological operations are used to improve the connectivity of the candidate plate region. Let A denote the binary foreground set and B denote the structuring element. The dilation and erosion operations are defined as

$$A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}, \quad (8)$$

$$A \ominus B = \{z \mid B_z \subseteq A\}. \quad (9)$$

The closing operation is a dilation followed by an erosion:

$$A \bullet B = (A \oplus B) \ominus B. \quad (10)$$

In this project, a rectangular structuring element is used because the license plate is also approximately rectangular. Morphological closing can connect broken blue regions, fill small holes inside the plate area and make the candidate plate contour more complete.

After morphological processing, contours are extracted from the binary mask. For each contour C_i , its bounding rectangle is represented by

$$R_i = (x_i, y_i, w_i, h_i), \quad (11)$$

where (x_i, y_i) is the upper-left coordinate, and w_i and h_i are the width and height of the rectangle. The area and aspect ratio are calculated as

$$A_i = w_i h_i, \quad (12)$$

$$r_i = \frac{w_i}{h_i}. \quad (13)$$

A candidate region is accepted when

$$A_i > A_{\min}, \quad r_{\min} < r_i < r_{\max}. \quad (14)$$

In the experiment, $r_{\min}$ and $r_{\max}$ were mainly set to 2.0 and 5.8, which is suitable for horizontal blue license plates. If multiple candidate regions satisfy the constraints, the region with the largest area is selected:

$$R^* = \arg \max_{R_i} A_i. \quad (15)$$

A small padding is then added around R^* to avoid cutting off the plate border or edge characters. This localized plate image is used as the input for the next stage.



Fig. 2 License plate localization result for Zhe A809JC



Fig. 3 License plate localization result for Zhe B2FS80

1.4 Plate Preprocessing and Binarization

The cropped plate image is converted to grayscale before character segmentation. The grayscale value $I(x, y)$ is calculated by a weighted sum of the three color channels:

$$I(x, y) = 0.299R(x, y) + 0.587G(x, y) + 0.114B(x, y). \quad (16)$$

This conversion reduces the computational complexity while preserving the contrast between the white characters and the blue background.

To suppress random noise, Gaussian filtering is applied to the grayscale image. The two-dimensional Gaussian kernel is defined as

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right), \quad (17)$$

and the filtered image is obtained by convolution:

$$I_g(x, y) = I(x, y) * G(x, y). \quad (18)$$

Here, σ controls the smoothing strength. A small Gaussian kernel is sufficient for this task because excessive smoothing may blur character edges.

After filtering, Otsu thresholding is used to convert the grayscale plate image into a binary image [4]. Let the gray-level histogram contain L levels. If a threshold t divides the pixels into foreground and background classes C_0 and C_1 , their probabilities are

$$\omega_0(t) = \sum_{i=0}^t p_i, \quad \omega_1(t) = \sum_{i=t+1}^{L-1} p_i, \quad (19)$$

where p_i is the probability of gray level i . The class means are

$$\mu_0(t) = \frac{1}{\omega_0(t)} \sum_{i=0}^t i p_i, \quad (20)$$

$$\mu_1(t) = \frac{1}{\omega_1(t)} \sum_{i=t+1}^{L-1} i p_i. \quad (21)$$

Otsu's method selects the optimal threshold t^* by maximizing the between-class variance:

$$t^* = \arg \max_t [\omega_0(t) \omega_1(t) (\mu_0(t) - \mu_1(t))^2]. \quad (22)$$

The binary image is then obtained by

$$B(x, y) = \begin{cases} 255, & I_g(x, y) \geq t^*, \\ 0, & I_g(x, y) < t^*. \end{cases} \quad (23)$$

Because the characters on blue plates are bright, the white foreground in the binary image corresponds

mainly to license plate characters. Finally, the plate border is removed by proportional cropping so that the upper and lower frame lines do not interfere with character extraction.



Fig. 4 Cropped plate image for Zhe A809JC



Fig. 5 Binary plate image after border removal

1.5 Character Segmentation

Character segmentation is performed on the binary plate image. Chinese provincial characters contain more strokes than English letters and digits. In a small training set, directly recognizing all Chinese provincial characters is unstable. Therefore, this project limits the target to Zhejiang plates and treats the first character as a fixed provincial prefix. In the segmentation step, the first character region is preserved by fixed-position cropping according to the layout of Chinese license plates. The remaining six alphanumeric characters are segmented by contour detection.

For each connected component or contour in the right part of the plate, a bounding rectangle is computed:

$$Q_j = (x_j, y_j, w_j, h_j). \quad (24)$$

The following geometric constraints are used to determine whether Q_j is a valid character region:

$$h_{\min} < h_j < h_{\max}, \quad (25)$$

$$w_{\min} < w_j < w_{\max}, \quad (26)$$

$$\rho_{\min} < \frac{w_j}{h_j} < \rho_{\max}. \quad (27)$$

These constraints remove small noise regions, screws and border fragments. The remaining candidate regions are sorted from left to right according to their horizontal coordinate:

$$x_1 < x_2 < \dots < x_6. \quad (28)$$

For normalization, each segmented character image is first cropped to its foreground bounding box and then resized while preserving its aspect ratio. If the

original character size is $w \times h$, the scale factor is

$$s = \min\left(\frac{W_n}{w}, \frac{H_n}{h}\right), \quad (29)$$

where $W_n = H_n = 20$ is the target image size. The resized character is placed at the center of a 20×20 black canvas. This normalization reduces the influence of different character widths and heights.



Fig. 6 Character segmentation result for Zhe A809JC



Fig. 7 Character segmentation result for Zhe B2FS80

1.6 PCA-SVM Character Recognition

The character recognition model is based on support vector machine classification [5]. First, digits 0–9 and capital letters A–Z were generated using computer fonts. Each class contained 80 synthetic samples, giving 2,880 synthetic samples in total. To reduce the domain gap between synthetic fonts and real license plate characters, the segmented real characters from the two tested plates were also added to the training set. For each real character, small rotation, scaling and translation transformations were used to create augmented samples.

Each normalized character image is flattened into a 400-dimensional vector:

$$\mathbf{x} = [p_1, p_2, \dots, p_{400}]^T, \quad (30)$$

where p_k is the normalized pixel value. Before classification, principal component analysis (PCA) is applied to reduce feature redundancy. Given n training samples, the mean vector is

$$\boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i, \quad (31)$$

and the covariance matrix is

$$\mathbf{C} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^T. \quad (32)$$

The principal components are obtained by solving

$$\mathbf{C}\mathbf{u}_k = \lambda_k \mathbf{u}_k, \quad (33)$$

where λ_k is the eigenvalue and $\mathbf{u}_k$ is the corresponding eigenvector. The feature vector is projected into a lower-dimensional subspace:

$$\mathbf{z} = \mathbf{U}_d^T(\mathbf{x} - \boldsymbol{\mu}), \quad (34)$$

where $\mathbf{U}_d$ contains the first d principal components. In this project, PCA is used before SVM training to reduce dimensionality and improve classification stability.

For binary SVM classification, the optimization objective is

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i, \quad (35)$$

subject to

$$y_i(\mathbf{w}^T \phi(\mathbf{z}_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad (36)$$

where $\phi(\cdot)$ is the nonlinear mapping function, C is the penalty parameter, and ξ_i is the slack variable. The radial basis function (RBF) kernel is used:

$$K(\mathbf{z}_i, \mathbf{z}_j) = \exp(-\gamma \|\mathbf{z}_i - \mathbf{z}_j\|^2). \quad (37)$$

The final decision function can be expressed as

$$f(\mathbf{z}) = \text{sign} \left(\sum_{i=1}^n \alpha_i y_i K(\mathbf{z}_i, \mathbf{z}) + b \right). \quad (38)$$

For the multi-class recognition problem involving digits and capital letters, the SVM classifier is extended using a one-versus-one strategy. Compared with simple template matching, the PCA-SVM classifier can better tolerate small font differences, slight deformation and binarization noise.

2 Experiment

2.1 Experimental Data and Evaluation Method

Two Zhejiang blue license plate images were used to verify the complete pipeline. The first image contains the plate Zhe A809JC, and the second image contains the plate Zhe B2FS80. The experiment records the plate localization result, character segmentation result and final recognition output. Since the first Chinese character was fixed as Zhejiang, the evaluation mainly focused on the six alphanumeric characters after the

provincial prefix.

The plate-level recognition result was considered correct only when all six alphanumeric characters were correctly recognized. The character-level result was calculated by counting the number of correctly recognized alphanumeric characters. If N_c represents the number of correctly recognized characters and N_t represents the total number of tested characters, the character-level accuracy is defined as

$$Acc_c = \frac{N_c}{N_t} \times 100\%. \quad (39)$$

Similarly, if N_p represents the number of correctly recognized plates and N represents the total number of tested plates, the plate-level accuracy is defined as

$$Acc_p = \frac{N_p}{N} \times 100\%. \quad (40)$$

2.2 Experimental Results

The final recognition results are shown in Fig. 8. The corresponding recognition statistics are listed in Tab. 1. Both tested Zhejiang license plates were correctly localized, segmented and recognized.

The two test images were both correctly recognized. The plate-level recognition accuracy on this small test set was 2/2, and the alphanumeric character-level recognition accuracy was 12/12. According to Eq. (31) and Eq. (32), the character-level recognition accuracy and the plate-level recognition accuracy in this limited test set both reached 100%.

During the early experiment, errors such as B being recognized as 8, F as E and S as 5 appeared because the original SVM model was trained mainly on synthetic font images. After real segmented characters were added to the training set and augmented, these confusions were reduced. This result shows that real samples are important when the training data and actual license plate fonts are different.

However, the current experiment is still limited. First, the system is designed for blue Zhejiang plates and does not automatically recognize provincial Chinese characters. Second, only two images were tested, so the reported recognition rate cannot represent general performance. Third, the color-based localization method may be affected by extreme illumination, motion blur, night scenes, green new-energy plates or serious plate inclination. These limitations should be improved in future work by collecting more samples, adding Chinese character templates or training a larger recognition model.



(a) Zhe A809JC



(b) Zhe B2FS80

Fig. 8 Final recognition results of the two Zhejiang license plates

Tab. 1 Experimental recognition results

Image	Ground truth	Recognized result	Correct
car2.jpg	Zhe A809JC	Zhe A809JC	Yes
car3.jpg	Zhe B2FS80	Zhe B2FS80	Yes
Total	2 plates	2 plates	2/2

3 Conclusion

This paper designed and implemented a machine-vision-based recognition method for Zhejiang blue license plates. The system uses HSV color segmentation and morphological processing to locate the blue license plate, applies grayscale conversion, Gaussian filtering and Otsu binarization to preprocess the plate, segments characters with a fixed provincial region and contour filtering, and recognizes the six alphanumeric characters by a PCA-SVM classifier. In the method design, HSV thresholding is used to construct the plate mask, morphological closing improves the connectivity of the plate region, geometric contour constraints locate the candidate plate, and PCA-SVM completes character recognition in a reduced-dimensional feature space. Synthetic samples and real augmented samples are combined for training, which improves the recognition of actual plate fonts. The two tested Zhejiang plates, Zhe A809JC and Zhe B2FS80, were correctly recognized. The project demonstrates a complete and reproducible traditional machine vision pipeline for a license plate recognition course assignment. Future work should expand the dataset, support more provincial Chinese characters and improve robustness under complex lighting and viewing conditions.

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